INCH POUND

MIL-S-29461
07 April 1992

MILITARY SPECIFICATION

SUIT, CHEMICAL AND BIOLOGICAL PROTECTIVE (CARBON SPHERE)

This specification is approved for use by US Marine Corps, Department of the Navy, and is available for use by all Departments agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a chemical and biological protective suit (coat and trouser).

1.2 Classification. The chemical and biological protective suit shall be of two-piece design, to include a coat, with built-in hood, and trousers.

Class 1. - Woodland Camouflage Printed
Class 2. - Desert Camouflage Printed

Schedule of Sizes

The chemical and biological protective suit shall be furnished in the following sizes, and as specified in Section 6.2:

Small
Medium
Large
X-Large

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Clothing and Textiles Directorate, Attn: DPSC-FSSD, 2800 South 20th Street, Philadelphia, PA 19101-8419, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 8415

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2. APPLICABLE DOCUMENTS

2.1 Government Documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

V-F-106	-	Fasteners, Slide Interlocking
TT-I-1795	-	Ink, Marking, Stencil, Opaque (porous and non porous Surfaces)
DDD-L-20	-	Label: for Clothing, Equipage, and Tentage (General Use).
PPP-B-636	-	Box, Shipping, Fiberboard

MILITARY

MIL-B-117	-	Bags, Sleeves and Tubing-Interior Packaging
MIL-F-21840	-	Fastener Tapes, Hook and Pile, Synthetic
MIL-C-43303	-	Cord, Elastic, Cotton
MIL-T-43548	-	Thread, Polyester, Cotton-covered and Rayon Covered
MIL-T-43566	-	Tape, Textile, Cotton, General Purpose, Natural or in Colors
MIL-C-29462	-	Cloth, Laminated, Polyester Tricot Knit, Activated Carbon Sphere, Chemical Protective
MIL-C-43468	-	Cloth, Camouflage Pattern: Wind Resistant Poplin, Cotton.

STANDARDS

FEDERAL

FED-STD-101	-	Test Procedures for Packaging Materials
FED-STD-595	-	Colors
FED-STD-751	-	Stitches, Seams and Stitchings

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	-	Marking for Shipment and Storage
MIL-STD-147	-	Palletized Unit Loads
MIL-STD-1668	-	Provisions for Evaluating Quality of Cloth Coveralls

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(Unless otherwise indicated, copies of federal and military specifications and standards are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

5-10-47 - Fastener, 1 inch
1-6-102 - Slide, 1 inch

(Application for copies maybe be requested from the Commander, Natick Research Development and Engineering Center, Kansas Street, Natick, MA. 01760).

TEST METHODS

CRDC-SP-84010 Laboratory Methods for Evaluating Protective Clothing Systems Against Chemical Agents.

(Application for copies maybe be requested from the Commander, Chemical Research and Development Center, ATTN: DRSMC-CLJ-IR (A), Aberdeen Proving Ground, Maryland 21010.)

2.2 Non-Government Publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation.

ANNUAL BOOK OF ASTM STANDARDS

ASTM D. 3951 Practice for Commercial Packaging

(Applications for copies of the ASTM Annual Book of ASTM Standards should be addressed to: ASTM, 1916 Race Street, Philadelphia, PA 19103-1187.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Standard sample. The Chemical and Biological Protective Suit shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6.3).

3.2 First article. Unless otherwise specified (see 6.2), before production is commenced, finished suits shall be submitted, or made available to the contracting officer or his authorized representative for approval. The approval of the first article authorizes the commencement of production but does not relieve the contractor of responsibility for compliance with all applicable provisions of the specification. The first article shall be manufactured by the contractor in the same facilities to be used for the manufacture of the production items.

3.3 Material.

3.3.1 Outer shell. The camouflage printed and quarpel treated cotton cloth used for the outer shell shall conform to the requirements specified in MIL-C-43468 and Section 6.2.

3.3.2 Lining. The laminated cloth used for the lining shall conform to MIL-C-29462.

3.3.3 Elastic cord. The cotton elastic cord for the drawcord at bottom of coat and for the hood drawcord shall be 3/16 inch in diameter, conforming to class 2 of MIL-C-43303. The color shall be camouflage green #483 for Class I and tan #380 for Class II, and the colorfastness requirements shall not apply.

3.3.3.1 Elastic cord resin impregnating agent. The resin agent used for dipping or impregnating the ends of the drawcords shall be cellulose acetate or cellulose acetate butyrate to prevent ravelling. The dipped or impregnated end shall be at least 1/2 inch in length.

3.3.4 Slide fasteners. The side fasteners shall be continuous element nylon or polyester (spiral type), size MS, type IV, style 8, and conforming to the requirements of V-F-106, except that the salt spray test shall not apply. The closed-end bottom stop shall be a staple type and shall be positioned and sewn, enclosing the sides and ends of the fastener chain. Solid type closed-end bottom stops shall abut the end of the slide fastener tape without damage to the tape or chain. The slide fastener pulls shall be 1" and shall have a hole with sufficient size to accommodate a 3/8 inch wide thong.

3.3.4.1 Slide fastener tape. The color of the slide fastener tape shall be #ODS66519 for Class I and #CA66516 for Class II, and the colorfastness requirements shall not apply.

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3.3.5 Hook and pile fastener tape. The hook and pile tapes for the trouser shall be 1" wide for the waist adjustment and for the leg adjustment it shall be 1" pile and 1 1/2" wide hook. The hook and pile fastener tapes for securing coat and trouser pockets and flaps, and coat center front overlap shall be 5/8 inch wide hook and pile tape with the exception of coat pocket which should be 1" pile, and for sleeve cuff adjustments shall be 1" inch wide hook and pile tape conforming to type II, class 1 of MIL-F-21840. The color shall be camouflage green #483 for Class I and tan #380 for Class 2,, and the colorfastness requirements shall not apply.

3.3.6 Suspender hardware.

3.3.6.1 Fastener (side release). The fastener (side release) shall meet the requirements of U.S. Army Natick Research Development and Engineering Center Drawing 5-10-47. The fastener shall be made from Black Acetal Delrin plastic (See 6.7).

3.3.6.2 Slide (triglide). The slide (triglide) shall meet the requirements of the U.S Army Natrick Research Development and Engineering Center Drawing Number 2-6-102. The slide shall be made from Black Acetal Delrin plastic (See 6.7).

3.3.7 Tape

3.3.7.1 Suspender strap tapes. The suspender strap tape shall be type I, class 8, one inch width. The color shall be camouflage green #483 for Class I and tan #380 for Class II conforming to MIL-T-43566 except that the colorfastness requirements shall not apply.

3.3.8 Thread.

3.3.8.1 Cotton or rayon wrapped polyester core thread. The cotton or rayon wrapped polyester core threads for seaming and stitching the coat and trousers shall be ticket Nos. 30, 50 and 70, 2 ply conforming to MIL-T-43548.

3.3.8.2 Color. The thread color shall be camouflage green #483 (color chip #34094 of FED-STD-595) for Class I and khaki P-1, cable #66019, for Class II and the colorfastness requirements shall not apply. The thread used to sew the laminate cloth for the lining may be white where not exposed to outside.

3.3.8.3 Water repellant treatment. The cotton or rayon polyester core thread shall be water repellant treated conforming to type II, class 3, of MIL-T-3530. Lubrication of the quarpel treated thread by any means in the end item fabricator's plant prior to or during the sewing is prohibited.

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3.3.9 Labels. All clothing labels shall be in accordance with DDD-L-20. Each pair of trousers and coat shall have an identification, size and instruction label. These three labels may be combined. The color shall be white and the colorfastness requirements shall not apply.

3.3.9.1 Identification label. The identification label shall conform to type VI, class 1 of DDD-L-20, except that the surveillance marking requirements of the laminated cloth shall appear near or under the last line of inscription on the label. The surveillance marking may be hand printed or stamped. Size of characters for the lot number shall be a minimum of 10 points affixed with Black (No. 37038) ink conforming to type I of TT-I-1795.

3.3.9.2 Size label. The size label shall conform to type VI, class 2 of DDD-L-20.

3.3.9.3 Instruction label. The instruction label shall conform to type VI, class 3 of DDD-L-20 and shall include the following instructive information for each respective size. Instructive information shall be printed in letters not less than a minimum 10 point (approximately 1/8 inch). The instruction label shall be centered below the combination size and contractor's label and stitched on all four sides.

Chemical and Biological Protective Suit (Carbon Sphere)

1. Wear as outergarment layer of clothing for protection against chemical agents.

Number 2 shall be as follows for each separate size.

2. The multisize system permits the size small to be worn by an individual with height/chest measurements of 5'3"/36".

2. The multisize system permits the size medium to be worn by an individual with height/chest measurements of 5'7"/39".

2. The multisize system permits the size large to be worn by an individual with height/chest measurements of 5'11"/39".

2. The multisize systems permits the size extra large to be worn by an individual with height/chest measurements of 6'3"/42".

3. **DO NOT REMOVE THIS LABEL**

3.3.9.4 Combination label. Size, identification and instruction labels may be combined conforming to type VI, class 14 of DDD-L-20 except that the lot number of the laminated cloth shall appear under the last line of inscriptions on the label and may be hand printed or stamped. Size of characters for the lot number shall be a minimum of 10 points affixed with Black (No. 37038) ink conforming to type I of TT-I-1795.

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3.4 Design.

3.4.1 Coat. The coat shall be of a hip length design with integrated hood and full length sleeves with adjustable tab and fastener tapes. The coat shall have two breast pockets, a slide fastener front closure with full length fastener tapes and elastic drawcords in the hem and hood.

3.4.2 Trousers. The trousers shall have a gusset on each side of the waist, two bellow type pockets with flaps located at the side, waist adjustment straps and suspenders. Each leg shall have an adjustment tab at the outside to control the adjustment opening.

3.5 Patterns. Standard patterns, which provide 1/2 ± 1/8 inch seam allowance for double lapped and double stitched seams of coat and trousers, and 3/8 ± 1/8 inch seam allowance for single needle seams, except where otherwise specified, will be furnished by the Government. The standard patterns shall not be altered in any way and are to be used as a guide for cutting the contractor's working patterns. The working patterns shall be identical to the standard patterns.

3.5.1 List of pattern parts. The components of the suit shall be cut from the materials as specified in accordance with the pattern parts indicated in Table I.

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Table I. Required Patterns

Material	Nomenclature of parts	Cut parts
	COAT	
Outer shell MIL-C-43468	Top sleeve elbow patch	2
	Under sleeve elbow patch	2
	Front	2
	Back	1
	Hood center panel	1
	Hood side panel	2
	Top sleeve	2
	Under sleeve	2
	Chest pocket	2
	Breast pocket	2
	Breast pocket flap	2
	Sleeve tab	2
Lining MIL-C-29462	Front	2
	Back	1
	Hood side panel	2
	Top sleeve	2
	Under sleeve	2
	Hood center panel	1

Material	Nomenclature of parts	Cut parts
	TROUSERS	
Outer shell MIL-C-43468	Side	2
	Cargo pocket facing	2
	Cargo pocket	2
	Cargo pocket flap	2
	Knee patch	2
	Gusset	2
	Waist tab	2
	Leg tab	4
	Bib	1
Lining MIL-C-29462	Side	2
	Gusset	2
	Bib	1

3.6 Construction.

3.6.1 Stitches, seams and stitching. Stitch, seam and stitching types specified in Table II shall conform to FED-STD-751. When two or more methods of seams or stitch types are given for the same operation, any one may be used. Where stitch type 401 is used, the looper, that is the underthread, shall be on the inside of the coat and trousers. The gage for double lapped and double stitched seams shall be 1/4 to 5/16".

3.6.2 Stitches per inch. The minimum and maximum number of stitches per inch shall be as specified in Table II.

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3.6.3 Thread breaks and ends of seams. Ends of all seams and stitching produced with 301 stitch type, when not caught in other seams or stitching, shall be backtacked not less than 1/4 inch. Thread breaks, all stitch types, shall be secured by stitching back of the break, not less than 1/2 inch. Skipped stitches or thread breaks, 401 stitch type, may be repaired using 301 stitch type. Ends of a continuous line of stitching shall overlap not less than 1/2 inch.

3.6.4 Eyelets. Eyelets, when used, shall have an inside diameter of 3/16 to 1/4 inch. The bight of the eyelet stitch shall be not less than 1/16 inch, the purling shall be on the outside and there shall be at least four overlapping stitches.

3.6.5 Mends. The laminate cloth used for the lining may be mended. The mends shall be made by pinching the material face to face where damaged, and stitching from 1/16 to 1/8 inch from end to end, to form a dart, tapered on each end. The length of the mend shall not exceed 1-1/4 inches. The following repair methods are acceptable providing that the contractor performs 100 percent in-process examination of the lining and shall for the repairable defects herein. Repair procedures and limitations that apply are as follows:

1. Repairs to open seams present in the outer shell of coat or trousers may be made by sewing through the shell and lining. The length of the repair stitching shall not exceed 2 inches. Repair stitching shall conform to 3.6.1 through 3.6.3, as applicable. Repairs are limited to two on the coat and two on the trouser.

2. Repairs to the lining may be hand sewn when plies of material are too thick to be machine sewn or the repairable defect is inaccessible by machine. The thread and stitches per inch shall be as specified in table II for the applicable operation.

3. Repairs on open seams in the joining seams of the lining shall be made by pinching the open seam together and stitching both plies 1/16 to 1/8 inch from end to end to form a dart tapered on each end. Shell material shall not be caught in the stitching and length of the repair shall not exceed 2 inches. Repairs are limited to two on the coat and two on the trouser.

4. Where there are more than two defects on the lining or the shell of any coat or trousers or when it is necessary to repair the shell or lining for an open seam in excess of 2 inches, the hem shall be opened and the repair made on the inside in conformance to the applicable operation in the Table II of the document.

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3.6.6 Bartacking. Bartacking shall be 1/2 ± 1/16 inch long, 1/8 ± 1/32 inch wide and shall contain 28 stitches. Bartacks shall be free from thread breaks and loose stitching.

3.7 Manufacturing operations requirements. The coat and trousers shall be manufactured in accordance with operations listed in Table II. Although the contractor is not required to follow the exact sequence of operations, the contractor is required to stitch the identification labels with surveillance marking already hand printed or stamped (see 3.3.9.1) to the lining (as specified in operation 6 of Table II) immediately upon cutting to ensure correct identification and traceability of the cloth. Any basting or holding stitching may be used providing threads or needle holes are not visible on the finished garments except for basting of suspenders.

3.8 Finished measurements. The finished measurements for coat and trousers shall be shown in Table III.

TABLE III. Finished measurements (inches)

	COAT		TROUSERS	
	1 /2 Chest 1 /	Back length 2/	Waist 3/	Inseam 4/
Small	24	28	37 3/4	30"
Medium	26	28 1/2	41 3/4	32"
Large	28	29	45 3/4	34"
X-Large	30	29 1/2	49 3/4	36"
Tolerance	±3/4	±3/4	± 1"	±1/2

1/ Measure across chest from folded side edge to folded side edge at base of armholes.

2/ Measure down center of back from collar joining seam to bottom edge of coat.

3/ Measure across front and back from seam and add measurements. Gussets are not included in measurement.

4/ Measure from crotch seam along inseam to bottom edge of leg.

3.9 Chemical agent protection. The chemical and biological protective suit shall provide chemical agent protection against the agent concentration levels and exposure time given in Table IV. (See 6.2)

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TABLE II- MANUFACTURING OPERATIONS REQUIREMENTS

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL BOB/ LPR
1.	CUTTING a) Cut the coat and trousers in strict accordance with patterns furnished, which show size, directional lines and marks for proper assembly. Unless otherwise specified, the directional lines indicate the warp direction. b) Cut the outer shell (cotton/ripstop cloth) parts from one piece of material except the leg tabs pocket facings, waist tabs, sleebe tabs which may be cut from ends. Parts cut from ends shall approximate the shade of the outer shell. Note: Pocket facings, leg tabs, sleeve tabs and waist tabs may be cut on rolls. c) Cut all parts of the lining (laminated cloth) from one lot of material. The knit side of laminated cloth is the face of the material. d) Directional lines on the patterns for front and back of trousers may vary from the warp direction by not more than 2-1/2 inches. Measurements shall be taken at the top and bottom edges of the pattern from the directional line to the selvage edge of the				

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL BOB/ LPR
1.	fabric and the difference between the two measurements shall not exceed 2-1/2 inches. e) Cut the elastic cord for the drawcord for the hood in 34 ± 1 inch lengths for all sizes. f) Cut the elastic cord for the drawcord at bottom of coat in the following lengths (+ inch tolerance) SIZE — INCHES Small — 54 Medium — 58 Large — 62 X-Large — 66 g) Cut the hook and pile portions of nylon fastener tape in the following lengths: COAT Breast pocket flap (2 each of hook) 5± 1/8 inches Breast pocket (2 each of pile) 5 ± 1/8 inches: Sleeve tab (2 each of hook) 4 1/2 ± 1/8 inches Sleeve hem (2 each of pile) 5 1/4 ± 1/8 inches Front, right pile entire length ± 1/2 inch Front, overlap, left hook entire length ± 1/2 inch				

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL BOB/ LPR
1.	TROUSERS Cargo pocket (2 each of pile) 8 1/2 ± 1/4 inches Cargo pocket flap (2 each of hook) 8 1/2 ± 1/4 inches Cut the one inch hook for waist adjustment straps (2 each of hook) 4 1/2 ± 1/4 inches for all sizes. Cut the one inch pile for waist adjustment (2 each of pile) 6 1/2 inches ± 1/4 inch for all sizes. Cut the one inch pile for leg adjustment (2 each of pile) 2 ± 1/8 inch. Cut the 1 1/2 inch pile for leg adjustment (2 each of pile) 3 1/4 ± 1/8 inch. Cut the 1 1/2 inch hook for leg adjustment strap (2 each of hook) 4 ± 1/4 inches. h) Cut the lengths of 1 inch cotton tape for the suspenders as follows: Front: 4 inch (cut 2) ± 1/2 inch Rear: 58 inch (cut 2) ± 2 inch				
2.	REPLACEMENT OF DAMAGED PARTS: Care shall be exercised during the spreading,				

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL	THREAD BOB/ LPR
2.	cutting and manufacturing operations to assure that material defects and damages, as classified in 4.4.3.2 are excluded and replaced with non-defective and properly matched material					
3.	MARKING: The outer shell parts of the suit (except the lining material) shall be marked, ticketed or bundled to insure a uniform size and shade throughout the garments. The lining material shall be bundled to insure a uniform size throughout the garments. Any method of marking the outer shell parts may be used except: a. Metal fastening devices b. Sew-on type markings c. Adhesive type tickets which discolor the mat material or leave traces of paper or adhesive upon removal of tickets.					
	MANUFACTURE COAT					
4.	MAKE HOOD a) Join hood side panels to hood center panel. Turn seam allowance to one side and raise stitch 3/16" to 1/4"	515 or 516 or 519 301	SSa-2 LSq-2	10/ 14 10/ 14	50 50	50 50

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL	THREAD BOB/ LPR
4.	b) Make a 3/16" to 1/4" diameter eyelet finished opening sewn through edge of hood as indicated by marks on pattern.	401 or 502 or 503		19-24 per eye-let	50	50
5.	JOIN SHOULDER SEAM OF OUTER SHELL: Join shoulder seams with a double lapped and double stitched seam with fronts overlapping backs.	301 or 401	LSc-2	10-14	50 50	50 70
6.	LABELS: Sew identification, size, and instruction label or combination label on all four sides to the knit side of lining on right front with bottom edges 2 + 1/4 inches from bottom edge of front and positioned as follows: (1) Size or combination label - centered (+ 1/2 inch tolerance) on right front. (2) Identification label (when used) positioned to left of size label with nearest side edges abutted. The nearest side edges may be overlapped and caught in one row of stitching. (3) Instruction label - positioned to the right of size label with	301	LSbj-1 and SSa-1	8-10	50	50

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL	THREAD BOB/ LPR
6.	nearest side edge abutted. The nearest side edges may be overlapped and caught in one row of stitching. NOTE: Final row of stitches on overlapped edges may be continued to bottom of lining.					
7.	JOIN SHOULDER SEAMS OF LINING: Position lining fronts and backs face to face and join shoulder seams.	301 or 401	SSa-1	8-10	50 50	50 70
8.	ATTACH HOOD Position hood to outer shell of coat, face to face, and join along neck edge from edge to edge.	301 or 401	SSa-1	10-14	50 50	50 70
9.	ATTACH ELBOW PATCHES TO OUTER SHELL. Position the under sleeve elbow patch on the wrong side of the under sleeve as indicated on the pattern and stitch 1/8 to 1/4 inch around edges. In a similar fashion position the top sleeve elbowpatch on the wrong side of the top outer shellsleeve as indicated on pattern and stitch 1/8 to 1/4 inch around the edge.	301	LSbj-1	10-14	50	50

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL	THREAD BOB/ LPR
10.	MAKE BREAST POCKET FLAP					
	a) Fold the breast pocket flap lengthwise and stitch a 1/4 inch seam on each end. Turn the flap forcing out the corners and place a row of stitching 1/4 to 5/16 inch on the closed sides of the flap and continue along the folded edge.	301 or 401 and 301	SSe-2(a) SSe-2(a) SSe-2(b) and OSf-1	10-14 10-14 10-14	50 50 50	50 70 50
	b) Position hook portion of nylon fastener tape on the breast pocket flap as indicated on pattern ±1/4 and stitch 1/8 inch to 3/16 inch from all edges.	301	LSbj-1	10-14	30	30
11.	MAKE BREAST POCKET					
	a) Turn top edge of pocket to inside, as indicated on the pattern and with the raw edge turned in, single stitch 1/16 to 1/8 inch from edge. The hem shall finish 1 1/4 ± 1/8 inches wide.	301 or 401	EFb-1	10-14	50 50	50 70
	b) Position pile portion of nylon fastener as indicated on pattern ± 1/4 and stitch 1/8 to 3/16 from all edges.	301 or 401	LSbj-1	10-14	30	30
	c) Stitch corners with two rows of stitching 1/16 to 1/8 inch apart using a 1/2 inch seam.	301	SSa-2	10-14	50	50

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL	THREAD BOB/ LPR
11.	d) As indicated on pattern fold both the side and bottom with the wrong side together and raise stitch 1/8 to 3/16 from edges. Fold face to face to form bellows on the side and bottom and raise stitch 1/8 to 3/16 inch along the side bottom.	301 or 401	SSa-2	10-14	50 50	50 70
12.	ATTACH BREAST POCKET AND FLAP TO OUTER SHELL a) Position breast pocket on the outer shell front as indicated on the pattern with the bellows towards the rear and turning in the raw edges stitch along the edges with a 1/8 inch raised stitch.	301	LSd-1	10-14	50	50
	b) Position raw edges of flap above pocket opening as indicated on pattern and join to front 1/8 to 3/16 inch from raw edges. Turn flap down and raise stitch 3/16 to 1/4 inch from turned edge, burying raw edges within the seam.	301	LSbl-2	10-14	50	50
13.	JOIN SLEEVES OUTER SHELL a) Join the top sleeve outer shell to the undersleeve outershell with a double lapped, double stitched seam. The top sleeve shall overlap the undersleeve.	301 or 401	LSc-2	10-14	50	50

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TABLE II- MANUFACTURING OPERATIONS REQUIREMENTS

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL	THREAD BOB/ LPR
14.	SET OUTER SHEEL SLEEVES a) Join the outer shell sleeve to the outer shell coat with a double lapped, double stitched seam. The coat shall overlap the sleeve.	301 or 401	LSc-2	10-14	50	50
15.	SIDE SEAM OUTER SHELL a) In one operation, join the outer shell underarm and outer shell side body seams with a double lapped, double stitched seam. The front shall overlap the back.	301 or 401	LSc-2	10-14	50	50
16.	MAKE COAT LINING a) Position left and right hood lining panels to center hood lining panel face to face and join with a 1/2" ± 1/8 seam.	301 or 401	SSa-1	8-10	50	50
	b) Join sleeve lining to coat lining with 1/2" seam ± 1/8.	301 or 401	SSa-1	8-10	50	50
	c) In one operation, join the lining under arm and side body seams with a 1/2" ± 1/8" seam.	301 or 401	SSa-1	8-10	50	50
	d) Position hood lining to coat lining face to face with edges even and join with 1/2"± 1/8" seam.	301 or 401	SSa-1	8-10	50	50

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL	THREAD BOB/ LPR
16.	e) At the back arm, join the top sleeve lining to the undersleeve lining with a 1/2" ±1/8" seam.	301 or 401	SSa-1	8-10	50	50
17.	JOIN LINING TO OUTER SHELL					
	a) Position lining to outer shell, face to face with fronts and hood edges even and join plies along front edges with the stitching continued along entire hood edges.	301 or 401	SSe-2(a)	8-10	50	50
	b) Turn coat to right side out, force out edges, and edge stitch fronts 1/4 to 5/16 inch from front edges with the stitching continued around the entire hood edge.	301	SSe-2(b)	8-10	50	50
	c) Form tunnel for hood drawcord by placing a second row of stitching 1/2" to 3/4" from edge stitching around entire front opening of hood. The drawcord shall not be caught in the stitching.	301	SSae-3(b)	8-10	50	50
18.	FINISH SLEEVE BOTTOMS a) With lining even with the finished hem of the outer shell, turn in bottom edge of outer shell as indicated on pattern. Fold in raw	301	SS1-1	8-10	50	50

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18.	edges of outer shell 1/4 ± 1/16 inch and stitch inch and stitch hem completely around the bottom from the turned edge. The hem shall finish 1" ± 1/4".					
	b) Sew pile to bottom of sleeve by marks indicated on pattern and stitch 1/8 to 3/16" from edge. For extra large only, velcro may be moved off the seam 1" ± 3/8"	301	Lsbj-1	10-14	30	30
19.	MAKE AND ATTACH SLEEVE TAB					
	a) Make sleeve tab. Fold sleeve tab in half crosswise and stitch both long raw edges. Turn tab, forcing out corners and stitch around edges of tab. ("1/8 to 3/16 inch)	301	SSe-2(a)	10-14	50	50
	or					
	Fold under both long raw edges along the seam line. Fold tab in half crosswise and stitch 1/4 to 5/16 inch around the edges of the tab.	301	SSe-2(b)	10-14	50	50
		301	SSc-1	10-14	50	50
	b) Position hook tape on the tab as indicated on the pattern and stitch 1/8 to 3/16 inch around edges of tab.	301	LSbj-1	10-14	30	30

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL	THREAD BOB/ LPR
19.	c) Attach sleeve tab to sleeve bottom, with 1/4" seam allowance, by marks indicated on pattern.	301	SSq-3(a)	10-14	30	30
	d) Turn sleeve tab back over the seam and raise stitch 3/16" to 1/4" from turned edge.	301	SSq-3(b)	10-14	30	30
20.	FINISH BOTTOM OF COAT					
	a) Make two buttonholes or sewn eyelets at the bottom of coat as indicated on pattern, vertical tolerance + 1/4" - 1/8", horizontal tolerance ± 1/2.	401 or 502 or 503		42-46 per button hole	50	50
	b) Position the elastic drawcord on inside of coat with end exiting through eyelets and knot ends of drawcord.			19-24 per eyelet	50	50
	c) Turn up bottom edge of outer shell even with the lining and with the raw edge turned in stitch 1/8 to 3/16 inch from turned edge with the stitching continuing along the complete bottom. The hem shall finish 1 ± 1/4 inches from the bottom edge of the coat.	301	SS1-1	8-10	50	50

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL	BOB/ LPR
21.	ATTACH SLIDE FASTENER					
	a) Position the back edge of the slide fastener tape on the inside of the left front 2 1/4 ± 1/4 inches from the finished front edge, as indicated on the pattern, with top stop 1/2 to 5/8 inch from the top.					
	b) Starting at bottom of tape, stitch tape to coat 1/16 to 1/8 from back edge, to the top of tape.	301	LSbj-1	8-10	50	50
	c) Place a second row of stitching through tape and coat positioned 1/4 to 3/16 inch from the first row of stitching.	301	LSbj-1	8-10	50	50
	d) Position the back edge of the slide fastener tape (tape containing slider) on outside of right front 2 ± 1/4 inches from the finished front edge, with the top stop 1/2 to 5/8 inches from the top.					
	e) Starting at bottom of tape, stitch tape to coat 1/16 to 1/8 inch from back edge to the top of coat.	301	LSd-1 or LSbj-1	8-10	50	50
	f) Place a second row of stitching through tape and coat positioned 1/4 to 1/16 inch from the	301	LSd-1 or LSbj-1	8-10	50	50

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL	THREAD BOB/ LPR
21.	first row of stitching.					
22.	ATTACH PILE TAPE FASTENERS a) Position pile tape on outer shell side of coat right front as indicated on pattern and corresponding to hook tape on coat left front. Stitch 1/8 to 3/16 inch through all thicknesses entire length of coat front. NOTE: One splice per coat of nylon fastener tape is allowed.	301	LSbj-1	8-10	30	30
	b) Position hook tape on lining side coat on left front as indicated on pattern and corresponding to pile tape on coat right front. Stitch 1/8 to 3/16 inch through all thicknesses entire length of coat front. NOTE: One splice per coat of nylon fastener tape is allowed.	301	LSbj-1	8-10	30	30
23.	MANUFACTURE TROUSERS MAKE CARGO POCKET FLAP a) Fold cargo pocket flap lengthwise and stitch 1/4 ± 1/16 inch seam on each end.	301 or 401 and 301	SSe-2(a) SSe-2(a) SSe-2(b) and OSF-1	10-14 10-14 10-14 10-14	50 50 50 50	50 70 50 50

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL	THREAD BOB/ LPR
23.	Turn the flaps forcing out the corners and place a row of stitching 1/4 to 5/16 inch on the closed sides of the flap and continue along the folded edge. The raw edges shall be caught in the two rows of stitching when attaching the flap to the garment. b) Position hook portion of nylon fastener tape on the cargo pocket flap as indicated on pattern, and stitch 1/8 to 3/16 inch from all edges.	301	LSbj-1	10-14	30	30
24.	<u>MAKE CARGO POCKETS</u> a) Make one 3/16 to 1/4 inch sewn eyelet at bottom of each cargo pocket, positioned as indicated on patterns.	401 or 502 or 503		19-24 per eyelet 26-32 per eyelet 26-32 per eyelet	50 70 70	50 70 70
	b) Fold the two front pleats as the marks are indicated on pattern and stitch along the fold 1/8 to 3/16 inch from the edge.	301	OSf-1	10-14	50	50

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL	THREAD BOB/ LPR
24.	c) Fold pleats as indicated by marks on pattern, toward back of pocket and stitch across top and bottom of pleats to hold in position.	301 or	SSa-1	6-14	50	50
		101 or	SSa-1	6-14	50	
		401	SSa-1	6-14	50	70
	d) With top raw edges of facing and pocket turned under, stitch facing to top inside of pocket, 1/16 to 1/8 inch from edge. NOTE: This operation may be performed with automated folders using facing cut off rolls no more than one splice per facing.	301 or	LSk-2	10-14	50	50
		401	LSk-2	10-14	50	70
	Stitch the facing to the top of pocket, turn and edge stitch 1/16 to 1/8 inch from top edge.	301 or	SSe-2	10-14	50	50
		401 and	SSe-2	10-14	50	70
		301	SSe-2	10-14	50	50
	After either of options, turn in bottom raw edge of facing and stitch to pocket 3/32 to 1/8 inch from the raw edge.	301 or	LSk-2	10-14	50	50
		401		10-14	50	70
	e) Bartack horizontally across fold of pleats with 1/2 inch bartacks. The bartacks shall be not more than 1/8 inch from the lower row of facing stitching.	Bar-tack		28 per bar-tack	50	50

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL	THREAD BOB/ LPR
24.	f) Fold outside edge of bellows and stitch along folded edge (through facing 1/16 to 1/8 inch from folded edge). Place a 1/2 inch vertical bartack, 1/8 ± 1/16 inch from top of pocket with bartack superimposed on stitching.	301 and Bar-tack	OSf-1	10-14 28 per bar-tack	50 50	50 50
	g) Fold bellows pleat at bottom of pocket; fold back side seam allowance in line with folded edge of pleat. Stitch across bottoms of pleat catching side seam allowance in the stitching.	301 or 401	LSd-1	6-14	50 50	50 70
	h) Position pile portion of nylon fastener as indicated on pattern and stitch 1/8 to 3/16 inch from all sides.	301	LSbj-1	10-14	30	30
25.	ATTACH KNEE PATCH TO THE FRONT OUTER SHELL Position the knee patch on the wrong side of front outer shell as indicated on the pattern and stitch 1/4 to 1/8 inch from all edges.	301	LSbj-1	10-14	50	50

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL	THREAD BOB/ LPR
26.	ATTACH CARGO POCKETS AND FLAPS TO OUTER SHELL Finished appearance. The bellows side of the cargo pocket shall finish toward the back. Pockets shall be uniformly stitched and properly positioned with the top corners of pocket openings covered by the flap. The edges of the cargo pocket protective piece shall not extend beyond the edges of the bellows pocket and flap. a) Mark position of cargo pockets and flaps, as indicated on the patterns. b) Turn in the seam allowance on the bellows side of pocket. Stitch bellows side of pocket to trousers, positioned as marked, with one row of stitching 1/16 to 1/8 inch from edge. The end of the seam shall catch the turned under seam allowance at bottom of pocket, closing the bellows at bottom.	301	LSd-1	10-14	50	50

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL	THREAD BOB/ LPR
26.	c) Turn in the other side and remaining bottom seam allowance of pocket and stitch to trousers with one row of stitching 1/16 to 1/8 inch from edge to bellows side.	301	LSd-1	10-14	50	50
				28 per bar-tack	50	50
	NOTE: Operations 22.b and 22.c may be performed in one operation to attach the cargo pocket. When using the single operation, the back bottom corner of each cargo pocket shall be bartacked through all layers for not less than 1/4 inch, and continue to stitch up the back side, pivoting at the corner such that the crossover stitch is immediately after the backtack to preclude any weak spot in this area.					
	d) Position raw edges of flaps above pocket openings as indicated on pattern, and join to trouser 1/8 to 3/16 inch from raw edges. Turn flap down and raise stitch 3/16 to 1/4 inch from turned edge making sure to bury edges within seam.	301 or	LSbl-2(a)	10-14	50	50
		401 and	LSbl-2(a)	10-14	50	70
		301	LSbl-2(b)	10-14	50	50
	or Turn in top raw edges of flap, positioned on trousers as indicated on pattern, and doublestitch	301	LSd-2	10-14	50	50

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL	THREAD BOB/ LPR
26.	burying raw edges within seam.					
	e) Bartack lower front corner of pocket flap with a 1/2" vertical bartack 1/16 to 1/8 inc. from side and bottom edges thru pocket flap, pocket and trouser.	Bar-tack		28 per Bar-tack	50	50
	f) Bartack back edge of cargo pocket opening with a vertical bartack in line with pocket stitching through pocket and trouser.	Bar-tack		28 per Bar-tack	50	50
27.	JOIN CROTCH, SEAT AND INSEAMS OF OUTER SHELL					
	a) Position left front at crotch and join the crotch seam beginning at the inseam and continuing to waist with a double lapped and double stitched seam.	301 or 401	LSc-2	10-14	50 50	50 70
	b) Join seat seam of outer shell with double lapped and double stitched seam. The seat seam may lap either direction.	401	LSc-2	10-14	30	50
	c) Join inseams with a double lapped and double stitched seam and the front will overlap the back.	301 or 401	LSc-2	10-14	50 50	70 50

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL	THREAD BOB/ LPR
28.	MAKE TROUSER LINING					
	a) Join crotch seam, seat seam, and inseam of trouser lining, stitch all seams with material face to face.	301 or 401	SSa-1	8-10	50 50	50 70
	b) Position gusset lining face to face with trouser lining and join on one side with 1/2 inch ± 1/8 inch seam. Join other side of gusset from top of waist down catching first seam at bottom of gusset and continued on 4 inches ± 1/2".	301 or 401	SSa-1	8-10	50 50	50 70
29.	ATTACH LABEL					
	Sew identification, size (combination identification, size and instruction label when used) and instruction label on all four sides to the knit side of left back of lining material (positioned as indicated below) with the top edge 3-1/8 ± 1/8 inches below top edge.	301	LSbj-1	8-10	50	50
	1) Size or combination label centered (± 1/2 inch tolerance) on left back.					
	2) Identification label (when used) - positioned to left of size label with nearest side edges abutted. The nearest					

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL	BOB/ LPR
29.	side edges may be overlapped and caught in one row of stitching. 3) Instruction label - positioned to the right of size label with nearest side edges abutted. The nearest side edges may be overlapped and caught in one row of stitching. NOTE: Final row of stitches on overlapped edges may be continued to top of lining.					
30.	SEW GUSSET TO OUTER SHELL Position gusset face to face with outer shell and join on one side from waist to bottom of gusset. Join other side of gusset from top of waist, catching first seam at bottom of gusset and continued on 4 inches ± 1/2.	301	SSa-1	10-14	50	50
31.	MAKE AND ATTACH WAIST TAB a) Make waist tab. Fold waist tab in half crosswise and stitch both long raw edges. Turn tab, forcing out corners and stitch around edges of tab. ("1/8 to 3/16 inch) or	301 301	SSe-2(a) SSe-2(b)	10-14 10-14	50 50	50 50

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL	THREAD BOB/ LPR
31.	Fold under both long raw edges along the seam line. Fold tab in half crosswise and stitch 1/4 to 5/16 inch around the edges of the tab.	301	SSc-1	10-14	50	50
	b) Position hook tape on the tab as indicated on the pattern and stitch 1/8 to 3/16 inch around edges of tab.	301	LSbj-1	10-14	30	30
	c) Attach waist tab to trouser back, with 1/4" seam allowance, by marks indicated on pattern.	301	SSq-3(a)	10-14	30	30
	d) Turn waist tab back over the seam and raise stitch 3/16" to 1/4" from turned edge.	301	SSq-3(b)	10-14	30	30
32.	SEW PILE TO TROUSER					
	a) Sew a piece of one inch wide pile to top of trouser for waist adjustment. Position pile by marks indicated on pattern and stitch 1/8" to 3/16" from all edges.	301	LSbj-1	10-14	30	30
	b) Position a piece of one inch wide and a piece of 1-1/2 inch wide pile to bottom of trousers leg for adjustment. Position pile by marks indicated on pattern and stitch 1/8 to 3/16 inch from all edges.	301	SSa-1	10-14	30	30

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL	THREAD BOB/ LPR
33.	MAKE AND ATTACH LEG TAB					
	a) Make leg tab. Fold leg tab in half crosswise and stitch both long raw edges. Turn tab, forcing out corners and stitch around edges of tab. ("1/8 to 3/16 inch)	301	SSe-2(a)	10-14	50	50
		301	SSe-2(b)	10-14	50	50
	or					
	Fold under both long raw edges along the seam line. Fold tab in half crosswise and stitch 1/4 to 5/16 inch around the edges of the tab.	301	SSc-1	10-14	50	50
	b) Position hook tape on the tab as indicated on the pattern and stitch 1/8 to 3/16 inch around edges of tab.	301	LSbj-1	10-14	30	30
	c) Attach leg tab to trouser with 1/4" seam allowance, by marks indicated on pattern.	301	SSq-3(a)	10-14	30	30
	d) Turn leg tab back over the seam and raise stitch 3/16" to 1/4" from turned edge.	301	SSq-3(b)	10-14	30	30
34.	FINISH WAIST					
	a) Form a dart at the top of gusset approximately 1 1/2 inches long by marks on pattern.	301	SSa-1	10-14	50	50

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL	THREAD BOB/ LPR
34.	b) Overedge stitch all raw edges around entire waist. Note: This operation may be performed before joining crotch, seat, and gusset seams or after.	502 or 503 504 505	EFd-1	6-10	50	70
	c) Turn in top edge of outer shell one inch as indicated on pattern (even with the lining material) and stitch 1/8 to 3/16 inches from the serged edge around entire waist. The finished waist hem shall be 3/4" ± 1/4 "	301	SSw-2(a)	10-14	50	50
35.	FINISH LEG BOTTOMS					
	a) Overedge stitch all raw edges around entire leg opening. Note: This operation may be performed before joining inseams or after.	502 or 503 504 505	EFd-1	6-10	50	70
	b) Turn in edge of outer shell one inch as indicated on pattern, (even with the lining material) and stitch 1/8 to 3/16" from the serged edge around entire leg. The finished hem shall be 3/4" ± 1/4".	301	SSw-2(a)	10-14	50	50
36.	MAKE BACK BIB AND ATTACH					
	a) Place lining bib on outershell bib and turn under edges of outer shell, 3/8 ± 1/8 inch, and hem around entire bib The finished hem should be 7/8" ± 1/8".	301	SSw-2(a)	10-14	50	50

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL	THREAD BOB/ LPR
36.	b) Join bib to back of trouser from side seam to sideseam. Turn in edges of bib even with side seam.	301	SSe-3(b)	10-14	30	30
	c) Bartack both ends of bib through bib and top of trouser, catching both rows of stitching in bartack.	Bar-tack		28 per Bar-tack	30	30
37.	MAKE SUSPENDERS AND ATTACH					
	a) Thread front suspender through female part of side release buckle. Align ends of strap and center a bartack 5/8" from folded end.	Bar-tack		28 stitches	50	50
	b) Place 3 bartacks in a triangular shape joining the rear suspenders 12" from the ends ± 1 inch. The short ends of the suspenders shall lay approximately 4 1/2" apart when laid flat and straight.	Bar-tack		28 stitches	50	50
	c) Place a 3/4" square boxtack centered over the three bartacks. All stitching should engage both plies of material. The boxtack should appear as a diamond shape enclosing the three bartacks.	301	SSa-1	10-14	50	50

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL	THREAD BOB/ LPR
37.	d) Thread long end of rear suspender through slide adjustment buckle, into male portion of side release buckle and back around bottom center of slide adjustment buckle. The raw edge end of the strap shall finish between the two straps. Center a 1/2" bartack approximately 1" from the slide adjustment buckle joining the end of the strap to the lower strap. Approximately 1 to 2" of the end of the strap shall extend below the bartack.	Bar-tack		28 stitches	50	50
	e) Join suspender with female portion of buckle to inside front of trouser with a 3/4" box-tack 1/8" from sides of strap, 3/8" from end of strap, and 1/8" from top of waist. All stitching shall be ± 1/16".	301	LSbe-2(b)	10-14	50	50
	f) Join suspender with male portion of buckle to inside back of trouser bib with a 3/4" boxtack 1/8" from sides of strap, 3/8" from end of strap and 1/8" from top of waist. All stitching shall be ± 1/16". End of strap will be turned back approximately 1 1/4" and stitching will be through two plies of material.	301	LSbe-2(b)	10-14	50	50

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STCH TYPE	SEAM/ STCH TYPE	STCH IN	THREAD NDL BOB/ LPR
37.	Turned end may be basted before attaching.				
38.	CLEAN COAT AND TROUSERS a) Trim end of threads to 1/4 inch maximum length. b) Remove all loose threads. c) Remove all size markings.				

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TABLE IV. Chemical Agent Protection Requirements

Test	Agent	Agent Concentration (minimum)	Exposure Time (minimum)
Liquid Vapor (emplaced drops)	HD	10 g/m^2	24 hours

3.10 Workmanship. The finish coat and trouser shall conform to the quality of product established by this specification. The occurrence of defects shall not exceed the applicable acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1. Responsibility for Inspection Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examination and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1. Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5.. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of insuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations is acceptable practice to ascertain compliance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Certificate of compliance. Where certificates of compliance are submitted, the Government reserves the right to check test such items to determine the validity of the certification.

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4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)
b. Quality conformance inspection (see 4.4)

4.3 First article inspection. The first article inspection shall consist of specified examinations and tests performed to determine that the production item conforms to the requirements of this specification.

4.3.1 First article samples. Unless otherwise specified, as soon as practical after the award of the contract or order, the manufacturer shall submit completed garments for subjection to the examination and tests specified in 4.3.2. The samples shall be representative of the construction, workmanship, components, and materials to be used during production.

4.3.2 First article tests. The first article sample shall be subjected to tests specified in 4.4.

4.3.2.1 Testing of first article suits. For first article testing, suits (see section 6.5) shall be submitted for all chemical agent protection testing in accordance with Table IV. Testing shall be performed as specified in 4.4.

4.4 Quality Conformance Inspection.

4.4.1 Quality Conformance Tests Unless otherwise specified sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.2 Component and material inspection In accordance with 4.1 above, components and materials shall be tested in accordance with all the requirements or referenced specifications, drawings, and standards, unless otherwise excluded, amended, modified or qualified in this specification or applicable purchase documents.

4.4.3. Examination of the end item Examination of the end item shall be in accordance with the provisions of 4.4.3.1 through 4.4.3.3.

4.4.3.1. Visual Examinations The end item shall be visually examined for critical defects listed in Table V. Any critical defect found during this examination shall be cause for the immediate rejection of the lot. The lot size shall be expressed in units of suits. The sample size shall be in accordance with inspection level II of MIL-STD-105.

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4.4.3.2 General Defects. General defects shall be classified as follows:

TABLE V - END ITEM VISUAL DEFECTS

DEFECTS	CLASSIFICATION		
	CRITICAL	MAJOR	MINOR
Material defects and damages			
a. Applicable to laminated lining cloth knit cloth side:			
1. Any cut, tear, needle chew, hole or burn thru the laminated cloth: 1/16 to 1/2 inch long inclusive		X	
-more than-1/2 long	X		
2. Mends in liner: -more than 1-1/4 inches in longest direction		X	
b. Applicable to cotton outer shell cloth:			
1. Any hole, needle chew, cut, tear, or burn		X	
2. Mends:			
- any mend up to 1/4 inch in diameter or length			X
- any mend more than 1/4 inch in any diameter or length		X	
- more than two mends			X
3. Any defect such as a smash, multiple float, or slub			X
4. Any misweave, area of poor dye penetration dye streak, broken or missing yarn, thin place or shade bar			X
c. Applicable to hook and pile fastener tape:			
- any cut, hole or tear			X
- any missing yarn			X
- width not as specified			X

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TABLE V - END ITEM VISUAL DEFECTS

DEFECTS	CLASSIFICATION		
	CRITICAL	MAJOR	MINOR
Shaded parts:			
Any outside part shaded except those cut from ends			X
Cleanness			
Spot or stain (grease, oil, ink)			X
Thread ends not trimmed or loose thread ends not removed from coat or trousers, to be scored only when the condition exists on major portion of the coat or trousers			X
Components and assembly			
Any component part or required operation omitted unless otherwise specified		X	
Any assembly or operation not properly performed, unless otherwise classified herein			X
Cutting			
Any component part not cut in accordance with directional lines indicated on pattern or not in accordance with specified requirements		X	
Seams and stitching			
a. Accuracy of seaming:			
1. Seams twisted, puckered, or pleated (unless otherwise classified herein)			X
2. Part of garment caught in unrelated operation of stitching			X

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TABLE V - END ITEM VISUAL DEFECTS

DEFECTS	CLASSIFICATION		
	CRITICAL	MAJOR	MINOR
3. Ends of seams or stitching produced with 301 stitch type, when not caught in other seams or stitchings, backtacked less than 1/4 or more than 1/2 inch			X
4. Stitching overlapping end(s) of thread breaks less than 1/2			X
5. Ends of a continuous line of stitching not overlapped, or overlapped less than 1/2			X
b. Open seam:			
1. Any on joining seam of laminate cloth except those listed in b.3 below:			
-up to a length of 1/4 inch inclusive			X
- over 1/4 but not greater than 1 inch in length inclusive		X	
- more than 1 inch in length	X		
2. Any seam except joining seam laminate cloth and those listed in b.3 below:			
-up to a length of 1/2 inch inclusive			X
-more than 1/2 inch in length		X	
3. Any on coat bottom, hem outside collar, trouser waistband turn under trouser leg bottom hem, sleeve hem, pocket stitching, slide fastener tape, or on fastener tape:			
-1/4 to 1/2 inch in length inclusive			x
-more than 1 inch		X	

"NOTE:
One or more broken stitches or two or more continuous skipped or runoff stitches on joining seam constitute and open seam. On double stitched seams, a seam is considered open when one or both sides of the seam is open."

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TABLE V - END ITEM VISUAL DEFECTS

DEFECTS	CLASSIFICATION		
	CRITICAL	MAJOR	MINOR
c. Raw edges (outer shell):			
1. On outside:			
- up to a length of 1/2 inch inclusive			X
-more than 1/2 inch in length		X	
NOTE: Raw edges not securely caught in stitching shall be classified as an open seam			
d. Runoffs:			
1. Joining seams: When resulting in an open seam, use "open seam" classification			
2. Edge or raised stitching (when not resulting in any open seam):			
-1/2 to 1 inch in length inclusive			x
-more than 1 inch		X	
e. Seam and stitch type:			
1. Wrong seam or stitch type			X
2. Looper thread of 401 stitch type finishing on outside of coat or trousers		X	
f. Stitch tension:			
1. Loose tension, up to 1 inch			X
2. Loose tension, more than 1 inch		X	
3. Tight tension (stitches break when normal strain is applied to the seam or stitching)		x	
NOTE: Puckering is evidence of tight tension. When puckering is evident, seam shall be tested by exerting pull in lengthwise direction of seam or stitching.			

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TABLE V - END ITEM VISUAL DEFECTS

DEFECTS	CLASSIFICATION		
	CRITICAL	MAJOR	MINOR
g. Stitches per inch (to be scored only when the condition exists on major portion of seam)			
1. less than specified:			
-one or two stitches			X
-more than two stitches		X	
2. One or more stitches more than specified			X
h. Stitches skipped or broken in any location other than those specified in 6.2 or 6.3.			
-up to a length of 1/2 inch inclusive			X
-over 1/2 inch but not greater than 1 inch in length		X	
-more than 1 inch in length	X		
Buttonholes or eyelets			
Not as specified			X
Not in specified position			X
Slide fastener			
a. Note: "The slide fastener shall be fully closed opened to determine if the slide fasteners operate smoothly and provide a secure closure." Following this defect, add the following defects: Any part of the slide fastener omitted, bent, broken, cracked of otherwise defective, affecting function.	X		
b. Bottom closed-end stop not attached as specified.		X	
c. Not specified type or size		X	
d. Fastener tape cut or torn		X	

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TABLE V - END ITEM VISUAL DEFECTS

DEFECTS	CLASSIFICATION		
	CRITICAL	MAJOR	MINOR
COAT/HOOD			
Elbow patches			
a. Top sleeve elbow patch missing		X	
b. Under sleeve elbow patch missing		X	
c. Top sleeve or under sleeve elbow patch not stitch as specified to the back side of the outer shell material		X	
Hood			
a. Elastic draw cord missing or caught in stitching	X		
Pockets and flaps:			
a. Edges of pockets pleated or twisted in stitching			X
b. Pocket flap tight causing fullness, twisting or curling of pocket flap			X
c. Bellows portion of breast pocket positioned toward front of pocket			X
d. Breast pocket or flap set on crookedly or poorly shaped			X
Slide fastener			
a. Top ends of tape not turned in and caught in the stitching joining slide fastener to coat			X

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TABLE V - END ITEM VISUAL DEFECTS

DEFECTS	CLASSIFICATION		
	CRITICAL	MAJOR	MINOR
b. Top stop more than 5/8 inch from neck edge or bottom edge of tape more than 1/4 inch from bottom of coat or extending beyond bottom edge			X
c. Back edge of slide fastener tape less than 1 3/4" or more than 2 1/4" from right coat front edge			X
d. Back edge of slide fastener tape less than 2" or more than 2-1/2" from left coat front edge			X
Sleeves			
a. Velcro sleeve tab set on crookedly			X
b. Velcro sleeve tab and pile tape out of alignment, irregular velcro sleeve tab or pile tape out of alignment by more than 1/2"		X	
c. Velcro sleeve tab and pile tape not able to be fastened	X		
-More than 3/8 inch and less than 3/4 inch			X
-3/4 inch or more		X	
Coat lining			
a. Knit side of laminate material not finishing toward the inside	X		
b. Seam allowance of joining seam not finishing toward the outer shell		X	
Labels			
a. Labels not positioned and attached to the knit side of right front lining as specified			X

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TABLE V - END ITEM VISUAL DEFECTS

DEFECTS	CLASSIFICATION		
	CRITICAL	MAJOR	MINOR
b. Size, instruction, identification or combination label (when applicable) missing, incorrect, or illegible			X
Bottom Hem			
a. Elastic drawcord missing or caught in the stitching	X		
TROUSERS			
Cargo pockets and flaps;			
a. Pleats turned toward front			X
b. Bellows side finished on front			X
c. Corner of flap caught in bartacks at end of bellows opening, restricting freedom of flap			X
d. Pocket flap tight causing fullness, twisting, or curling of pocket flap			X
e. Edge of pockets pleated or twisted in stitching			X
f. Bartack stitching loose, incomplete, broken, not fully engaging stitched material, or not as specified:			
-involving one bartack			X
-involving two or more bartacks		X	
Knee patches			
a. Knee patch missing		X	
b. Knee patch not stitched as specified to the backside of the outer shell material		X	
Trousers lining			
a. Knit side of laminate material not finishing toward the inside	X		

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TABLE V - END ITEM VISUAL DEFECTS

DEFECTS	CLASSIFICATION		
	CRITICAL	MAJOR	MINOR
b. Seam allowance of joining seams not finishing towards the outer shell		X	
Buckels			
a. Will not function or are missing	X		
Suspenders			
a. Suspenders missing	X		
Leg Tabs			
a. Tabs will not engage or are missing	X		
Waist Tabs			
a. Tabs will not engage or are missing		X	
Labels			
a. Label not positioned and attached to the knit side of left back lining, as specified			X
b. Size label, identification label, or combination label (when applicable) missing, incorrect, or illegible			X
c. Instruction label missing, incorrect, or illegible			X

4.4.3.3 Dimensional examination. The end item shall be examined for the defects listed below. The lot size shall be expressed in units of suits. The sample suit shall be one suit. The inspection level shall be in accordance with level II of MIL-STD-105.

DEFECTS

- Any measurement deviating from any dimension specified in table III
- Sleeve lengths uneven by more than 1/2 inch
- Leg lengths uneven by more than 1/2 inch

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4.4.4 Chemical agent protection. 1/ Compliance with chemical agent protection requirements of Table IV shall be determined on suits as shown in Table VI and is the responsibility of the agency placing the contract. At government contracting office request, QAR will ship the required number of suits to the agency placing the contract for chemical agent testing. The QAR will not cut any samples. The sample unit shall consist of one suit. Samples shall be cut from each garment and submitted for chemical agent testing. If any suits fail chemical agent testing, the government contracting office shall notify contractor, in writing, providing the test results. Failed suits shall be retested by an independent laboratory acceptable to the government and the contractor. Failure in initial testing shall not be cause for rejection if retest results are in compliance with the chemical agent protection requirements set forth in Table IV. Failure in initial testing and in such retest shall be cause for rejection of the lot from which the failed sample was drawn. The sample size shall be in accordance with the following:

Lot size (suits)	Sample size (each)
less than 5000	3
5001 and over	5

1/ For lot by lot evaluation, the Chemical and Biological Protective Suit shall be tested every five (5) lots or as directed by the procuring activity. Shipment of completed suits from the contractor will not be held awaiting results. Control of lot numbers will provide the necessary information required to recall a lot if chemical agent protection is not sufficient.

TABLE VI Chemical Agent Protection Test Methods

Test	Agent	Test Method	Reference
Liquid vapor (emplaced droplets)	HD	2.2	CRDC-SP-84010C

4.4.5 Examination of packaging requirements. The packaged items shall be examined as specified in MIL-STD-1668 unless otherwise specified.

4.4.6 Package examination. Each filled and sealed package shall be examined for defects listed below. The occurrence of any defect shall be cause for rejection of the package. Garments from rejected packages may be repackaged as specified in 5.1 and submitted for re-examination.

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DEFECTS

- o Closure seal width not as specified
- o Closure seal not continuous
- o Tear, cut or hole in package

4.4.7 Package leakage testing. Each package prior to packing, shall be left to stand a minimum of 4 hours after sealing. Following the 4 hour period each package shall be visually inspected for leakage. Any package that exhibits an increase in volume or soft hand where the vapor barrier bag is easily separated from the garment shall be considered a failure. Any failure shall be repacked and resubmitted for visual retesting.

4.4.8 In-process test of heat sealed closure of seam. The test shall be made daily prior to the start of filling and sealing packages to determine that the heat sealing equipment is producing seams in accordance with the requirements of 5.1.2. One full seam shall be sealed on any empty barrier bag using recommended heat seal settings and tested in accordance with method 2024 or FED-STD-101 at normal room temperature with a static load of 50 ounces using a fabricated bag specimen procedure. A 5 percent reduction in static load is permitted when room temperature in the test area exceeds 90°F. Any test failure shall be cause for readjustment of the sealer settings and retesting unit a satisfactory seam is produced.

5. **PACKAGING**

5.1 Preservation.

5.1.1 Folding. Each of the suit components (coat and trousers) shall have all fastenings secured and shall be neatly folded into a rectangle. All slide fasteners shall be located on the inside of the folded garment to prohibit contact with barrier bag. The coat and trouser shall be carefully rolled together with the trouser on the outside.

5.1.2 Sealing. The rolled suits shall be individually placed in a heat sealable, flexible, water vapor proof, grease proof barrier bag conforming to type I, class E, style 1 of MIL-B-117 - using MIL-B-131, class 1 tyvek exterior materials only. The bag shall be flat style with an approximate inner dimension of 15 X 23 inches and sealed on three sides with 1/2 $\pm$ 1/16 inch seal. Each bag shall have all air evacuated and be heat sealed while maintaining a vacuum of 18 to 24 inches of mercury. The final seal shall be a minimum of 3/16 inch wide.

5.1.3 Package marking. Each package shall have indelibly and legibly printed or stamped on it the correct garment nomenclature, size, stock number, date of packaging (month and year), and filter fabric lot number in letters no smaller than

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3/16 inch high. Additionally, the following information in letters no less than 1/4 inch (18 points) shall be indelibly and legibly printed or stamped: **WEAR AS OUTER GARMENT.**

5.1.4 Bagging. Every vacuum sealed barrier bag shall be inserted into a clear polyethylene pouch and heat sealed. The polyethylene pouch and heat sealed. The polyethylene pouch shall be fabricated from tubing or sheet stock with a minimum thickness of 0.004 inches and be provided with a 1/4 inch diameter hole near the bottom for excess air to escape.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Eight (8) to ten (10) suits of one size packaged as specified, shall be packed in a fiberboard shipping container assembled, closed and reinforced to conform to class weather-resistant,style RSC-L, grade V2s, size 3A of PPP-B-636. A liner conforming to type CF, Class Domestic Variety DW, Grade V15c of PPP-B-636 shall be placed inside carton. Packages shall be either packed flat and/or on end. All voids shall be filled with commercial grade kraft paper. The shipping container shall be sealed using glue. To accomodate the last shipment at the end of the contract, less than full cartons and pallets are acceptable.

5.2.2 Level B. Eight (8) to ten (10) suits of one size only, packaged as specified, shall be packed in a fiberboard shipping container assembled and closed in accordance with PPP-B-636. The container shall be size 3A, style RSC-L, Type CF (variety SW or SF), class Domestic, Grade 275 or PPP-B-636. A liner conforming to Type CF, Class Domestic, Variety DW, Grade 275 of PPP-B-636 shall be placed inside carton. Packages shall be either packed flat and/or on end. All voids shall be filled with commercial grade kraft paper. The shipping shall be contained using glue. To accomodate the last shipment at the end of the contract, less than full cartons and pallets are acceptable.

5.2.3 Level C. Suits packaged as specified shall be packed in accordance with the applicable requirements as ASTM D 3951-90.

5.2.4 Palletization. When specified (see 6.2), suits packed as specified in 5.2 shall be palletized on a 4-way entry pallet in accordance with load type of MIL-STD-147. Each prepared load shall be bonded with horizontal primary and secondary straps in accordance with bonding means C, or D or film bonding methods for F or G. Pallet pattern shall be in accordance with the appendix of MIL-STD-147. The pallet shall be 4-way, type I, class 1, style 1, Size A, wood group I, II, III,or IV of MIL-P-15011, or 4-way type IV, V or VIII, class 1, style A, size 2, wood group I, II, III, or IV, grade A of NN-P-71. Each pallet load shall contain the same number of cartons.

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5.3 Marking. In addition to any special marking required by the contract or purchase order, unit packs, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 and shall include the size of the suits, the date of packaging (month and year) and the filter fabric lot number.

5.3.1 Surveillance marking.

5.3.1.1 Unit pack. Each unit pack shall be marked with the surveillance marking indicated on the enclosed suit (see 3.3.9.1). The markings shall be printed or stamped below the identification and contract date markings in black characters.

5.3.1.2 Shipping container. Each shipping container shall be marked with the surveillance marking indicated with the surveillance marking indicated on the enclosed suit (see 3.3.9.1). The markings shall be stenciled, stamped or printed on the same side of the container as the identification and contract data markings and on one end of the container in black characters.

5.3.3 Special markings.

5.3.3.1 Unit pack. Each unit pack shall be marked in black with a capital letter "C" for class 1, Woodland camouflage, and a capital letter "D" for class 2, Desert Camouflage. This letter shall be minimum of 3 inches in height and located on the face of the vapor barrier bag.

5.3.3.2 Shipping container. Each shipping container shall be marked in black on three sides with either of the characters "8C", "9C" or "10C" for class 1 Woodland Camouflage suits and either "8D", "9D or "10D" for class 2 Desert Camouflage garments. These characters shall be minimum of 3 inches in height and centered on three sides of the shipping container. The side containing the identification and contract number need not be marked.

5.3.4 Labels, mixed sizes. Each shipping container packed with mixed sizes shall have securely attached to the end and side, directly under the printing or stenciling, a white paper label 5 by 4 inches in size, with the words "Mixed NSN's" plainly printed or stamped thereon and under these words shall be legibly printed or stamped the correct quantity and NSN's container therein.

5.3.5 Label, Mixed lots. Each shipping container packed with mixed lots shall have securely attached to the end and side, directly under the printing or stenciling and beside the label for mixed sizes, a white paper label 5 by 4 inches in size with the words "Mixed lots" plainly printed or stamped thereon and under these words shall be legibly printed or stamped the correct quantity and lot contained therein.

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6. **NOTES**

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The suit is intended for use by combat personnel to provide protection against battlefield concentrations of chemical agents in liquid and vapor form.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification
b. First article (see 3.2, and 4.2)
c. Type and size required (see 1.2)
d. Specified color of shell material and trim material (See 3.3.1 thru 3.3.9)
e. Requirements for agent testing, first article and production lots (See 3.9; 4.2; and 4.4.4)
f. Selection of applicable level of packaging (see 5.2)
g. Acceptance criteria

6.3 Standard sample. For information regarding the availability of standard sample, address inquiry to the procuring activity issuing the invitation for bids.

6.5 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52-209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.6 Acceptance criteria. The acceptance criteria below are recommended for use. The acceptance criteria as specified in the contract or purchase order shall be binding. Unless otherwise specified, the following acceptance criteria are in accordance with MIL-STD-105.

6.6.1 For end item visual examination. An acceptance quality level (AQL), expressed in terms of defects per hundred units, of 4.0 for major defects and 10.0 for total (major and minor combined) defects is recommended.

6.6.2 For end item dimensional examination. An AQL, expressed in terms of defects per hundred units, of 2.5 is recommended.

6.6.3 For packaging examination. An AQL, expressed in terms of defects per hundred units, 2.5 is recommended.

6.6.4 Examination of count in intermediate containers. An AQL, expressed in defects per hundred units, of 2.5 is recommended.

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6.6.5 Testing of the end item. An AQL, expressed in defects per hundred units, of 6.5 is recommended

6.7 Comparable Suggested Products. ITW Waterbury part number 101-0100 Black Acetal Delrin plastic meets the requirements of the fastener as specified in 3.3.6.1 ITW Waterbury part number 105-0100 Black ACetal Delrin plastic meets the requirements of the slide as specified in 3.3.6.2.

6.8 Subject term (key word) listing.

Cover
Protective

Custodian:
MC

User Activity:
MC

Review Activity:
MC

Preparing activity:
DLA - CT
Project No. 8410-0830

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8 In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-S-29461	2. DOCUMENT DATE (YYMMDD) 92-04-07

3. DOCUMENT TITLE
Suit, Chemical and Biological Protective (Carbon Sphere)

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY

a. NAME DLA -CT	b. TELEPHONE *(Include Area Code)* (1) Commercial: (215) 737-8105 (2) AUTOVON: 444-8105
c. ADDRESS *(Include Zip Code)* 2800 South 20th Street Philadelphia, PA 19101-8419	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

DD Form 1426, OCT 89 *Previous editions are obsolete.* 198-290

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